

AKHIL AGRAWAL

+91 9584293402 ♦ Mumbai, India

akhilagrawal14@gmail.com ♦ [linkedin](#) ♦ [github](#)

SUMMARY

Applied ML Engineer with 5+ years of experience building and shipping Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and NLP systems to production. Strong background in distributed systems, real-time ML infrastructure (Kafka, Pub/Sub), and API serving (FastAPI, Flask).

EXPERIENCE

Machine Learning Engineer – SDE 2

July 2024 - Present

Fynd (Shopsense Retail Technologies Ltd.)

Mumbai, India

- Served as **DRI** for GPU-based ML infrastructure on **GCP/Kubernetes**, owning 40+ production models across video super-resolution, image super-resolution, and AI background removal, resolving critical production incidents and maintaining service SLAs.
- Reduced GPU infrastructure costs by **28%** (approx. \$4,000/month) through dynamic scaling, model quantization (**TensorRT**, **torch.compile**), and resource optimization, with an additional **45.5%** cost reduction on Flux deployment by migrating from A100 to dual L4 GPU instances.
- Designed an intent-aware **prompt routing system** for image editing as a service, dynamically routing requests across self-hosted models, Replicate, Fal.ai, and Flux based on user intent and cost constraints, handling **1,000+ requests/day** across hosted services via **FastAPI** with async processing.
- Integrated **Kafka** and **Pub/Sub** for event-driven ML inference, decoupling services and reducing p95 processing latency by **20%** compared to the prior synchronous pipeline.
- Migrated machine learning workloads to **Ray** for distributed execution, reducing batch job processing time by **20%** and cutting associated compute costs by **10%**.
- Worked with DevOps, Backend, Growth, and Sales teams on product delivery and infrastructure decisions.

Machine Learning Engineer 2 / 3

July 2021 - July 2024

Comcast India Engineering Center

Chennai, India

- Built a production **semantic video search system** allowing ad-tech clients to query scenes using natural language, indexing **5,000+ hours** of content from NBC, Universal Studios, and Sky across TV series and movies using **CLIP**, an in-house model, vector database, and Elasticsearch, achieving 2-3s query latency and built to scale further.
- Built a **RAG-based LLM prototype** for sports information retrieval using **OpenAI Embeddings** for dense vector indexing with **Milvus** and summary indexing over captions and web sources, with **GPT-3.5** as the generation backbone, improving retrieval precision by 38% over a keyword-search baseline on internal evaluation, showcased to senior leadership and received strong endorsement for productionization.
- Owned end-to-end development of a **celebrity recognition framework** using **MTCNN** for detection and **FaceNet** embeddings with hybrid clustering, with a reference dataset of **12,000+ celebrities** achieving Precision@5 of **92%**, productionized and used by NBC, Sky, and other major broadcaster platforms.
- Built **OCR validation pipeline** for Xfinity TV to automatically assess voice query output quality, processing **8,000+ queries/day** with 94% agreement with human annotators and fully replacing manual QA review.

Summer Intern – Machine Learning

May 2020 - July 2020

Comcast India Engineering Center

Chennai, India

- Developed a computer vision and OCR pipeline to detect scoreboard patterns in cricket match videos, taken to production for Live Broadcasting by Sky.

EDUCATION

M Tech. in Computer Science and Engineering

Indian Statistical Institute, Kolkata

2019 - 2021

Aggregate: 84.9%

Bachelor of Engineering in Information Technology

Institute of Engineering And Technology, Indore

2015 - 2019

CGPA: 8.24/10

SKILLS

LLM / GenAI Programming	RAG, LangChain, Gemini API, Vector DBs (Milvus, FAISS), Prompt Engineering
Web Development	Python, R, C++, C
ML / DL Libraries	FastAPI, Flask, Node.js, Async (asyncio)
MLOps & Inference	PyTorch, PyTorch Lightning, Keras, TensorFlow, Scikit-Learn, ONNX, OpenVINO
Streaming & Infra	MLflow, DVC, ZenML, Ray, NVIDIA Triton, TensorRT, torch.compile, Quantization
Cloud Computing	Kafka, Pub/Sub, Docker, Kubernetes, Microservices, Distributed Systems
Data & Viz	GCP, AWS
	NumPy, Pandas, Matplotlib, OpenCV

PROJECTS

Cross-Domain Semantic Search using RAG and Gemini Embedding 2

- Built a personal knowledge search tool (similar to macOS Spotlight) using **Gemini Embedding 2**, supporting search across 5 content types: text files, PDFs, code, notes, and emails, indexing **5,000+ documents** across domains.
- Used hybrid retrieval (dense + sparse) with a re-ranking layer, achieving MRR of 0.81 and an 18% improvement over dense-only retrieval, served via **FastAPI** with sub-200ms query latency.

Multimodal Fake Content Detection using Facial Signals and Gemini

- Built a fake vs real detection system across 4 modalities (image, video, audio, text), extracting facial landmark and biometric signals using computer vision and passing structured features to **Gemini** for reasoning and final decision.
- Trained and evaluated on **FaceForensics++** (1,000 videos, 4 manipulation types) for video/image and **ASVspoof 2019** (25,000+ samples) for audio, with the text modality using an annotated dataset of 6,000 human vs AI-generated samples.
- Achieved **89%** accuracy on held-out test set, designed as a two-stage pipeline to keep decisions interpretable, applicable to KYC and fraud prevention.

Detecting Anemia from Retinal Images using Deep Learning

- Collaborated with Prof. Sushmita Mitra (ISI Kolkata) and Sankara Nethralaya, Chennai on a medical imaging research project.
- Experimented with CNN architectures (MobileNet V2, DenseNet 121, ResNet50), applied GradCAM for activation visualization and data augmentation to address class imbalance.
- Achieved AUC of 0.75 on a small, imbalanced clinical dataset (positive:negative ratio of 1:4).

INTERESTS AND ACHIEVEMENTS

- Excellence Award from Senior Leadership, Comcast, 2023.
- Qualified GATE CSE 2019, All India Rank **388** (99.66 percentile).